



Bipolar Disorder

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Introduction

Bipolar disorder is a complex, recurrent mood disorder that affects 1–2% of the population.^[1] It is defined as having experienced either a manic episode (with or without a history of major depressive episode) or a hypomanic episode (with a current or past history of major depressive episode). The *Diagnostic and Statistical Manual of Mental Disorders, 5th Edition* (DSM-5) criteria also require that mania or hypomania include abnormally and persistently increased goal-directed activity or energy (see [Table 1](#) and [Table 2](#)).^[2]

Symptoms of mania include changes in mood, energy, sleep requirements and ability to concentrate, and in some cases psychotic symptoms.

Symptoms of depression include oversleeping or profound tiredness (the most common symptom of bipolar depression), pessimism, inability to socialize, decreased cognitive abilities and possibly suicidal or psychotic symptoms.

Bipolar disorder is formally divided into several categories: bipolar I disorder, bipolar II disorder, bipolar disorder induced by medications/substances/other medical disorders, as well as categories for patients who do not meet usual criteria (see [Table 2](#)).^[2] Diagnosis of bipolar disorder is often difficult as the illness is perhaps the most complex psychiatric disorder. It has the most variable clinical presentation and is associated with the highest number of episodes, the highest degree of comorbidity and the highest mortality of the major psychiatric conditions. When bipolar disorder is severe and psychotic symptoms are present, it resembles schizophrenia. When accompanied by substance use, it mimics a pure substance use disorder. In a severe depression phase, it resembles unipolar major depression. When the symptoms are mild and changing rapidly, it resembles borderline personality disorder. In children and adolescents, it may resemble attention-deficit hyperactivity disorder. Moreover, differential diagnoses may be comorbidities, such as a manic patient who is also using cocaine, or quite commonly, a bipolar patient with a concomitant personality disorder.

Table 1: Description of Bipolar Disorder Episodes^[2]

Type of Episode	Features
Manic episode	Mood: Abnormally and persistently elevated, expansive or irritable; must have concomitant increase in activity or energy; psychotic symptoms may occur Duration: At least 1 wk, present most of the day, causing significant distress/disability or requiring hospital admission Plus: If mood is elevated or expansive, 3 or more of the following features must be present. If mood is predominantly irritable, 4 or more are required. • Grandiosity • More talkative • Excessive involvement in pleasurable activities that may have unpleasant consequences • Less need for sleep • Flight of ideas • Distractibility • More goal-directed activity
Hypomanic episode	Mood: Same symptoms as for mania, but milder and not disabling; <i>no</i> psychotic symptoms Duration: 4 days or longer and present most of the day
Major depressive episode	Mood: Depressed most of the day <i>or</i> markedly diminished interest or pleasure (anhedonia) Duration: At least 2 wk, with significant change from previous functioning Plus: Four or more of the following: • Insomnia or hypersomnia • Significant weight loss/gain or change in appetite • Fatigue or loss of energy • Psychomotor retardation or agitation (observable)

Type of Episode	Features
	• Worthlessness or excessive guilt • Impaired thinking, concentrating or making decisions • Recurrent thoughts of death, suicidal ideation or attempt/plan
Manic or hypomanic episode with mixed features	Criteria met <i>during same time period</i> for hypomanic episode or manic episode with 3 or more features of depressive episode
Depressive episode with mixed features	Criteria met <i>during same time period</i> for a major depressive episode with 3 or more features of manic or hypomanic episode

Table 2: Classification of Bipolar Disorder^[2]

Disorder	Features
Bipolar I Disorder	Lifetime history of at least 1 clear-cut manic episode with or without episodes of hypomania or depression
Bipolar II Disorder	History of hypomanic episode and major depressive episodes with no history of a full manic episode
Substance/Medication-Induced Bipolar and Related Disorder	Most diagnostic criteria for bipolar disorder met, but there is evidence that this developed during or soon after substance intoxication/withdrawal or medication use/withdrawal; the offending substance must be judged as capable of inducing such symptoms
Bipolar and Related Disorder due to Another Medical Condition	Most diagnostic criteria for bipolar disorder met, but there is evidence that this is the direct pathophysiological consequence of another medical condition
Other Specified (or unspecified) Bipolar and Related Disorders	Most diagnostic criteria for bipolar disorder met, but not all, e.g., duration of episode shorter than required for diagnosis, or history of hypomanic episodes but no depressive episodes that meet diagnostic criteria

Disorder	Features
Cyclothymic Disorder	Recurrent episodes of hypomania and mild (subthreshold) depressive symptoms

Goals of Therapy

- Control symptoms of acute episode
- Prevent recurrences
- Provide ancillary care for comorbid psychiatric conditions such as anxiety or substance use or medical conditions such as endocrine/metabolic disorders
- Restore optimal functioning and cognition

Investigations

- It is estimated that up to one-third of patients appearing in primary care settings with symptoms suggestive of unipolar major depression are actually experiencing depression in the context of bipolar illness.^[3] Ask all patients with depression about possible hypomanic or manic symptoms in their past. While no screening test is ideal, the [Mood Disorder Questionnaire \(MDQ\)](#) is a helpful, self-completed form that asks systematic yes/no questions about the symptoms of mania;^[4] an adolescent version also has been developed.^[5] Two or more “yes” answers on the MDQ should prompt a more thorough clinical review of symptoms, including questioning family/friends if possible.
- The diagnosis of bipolar disorder depends not just on the clinical presentation but, very importantly, on a reliable collateral history from a friend or family member who can corroborate episodes of elevated mood, inappropriate behaviour, decreased sleep with increased energy or grandiosity.
- It is critical to determine whether the patient has a *family history* of mood disorders, substance use or bipolar disorder itself.
- Laboratory investigations:
 - depend in part on the age and acute presentation of the patient
 - for all patients, basic blood tests include CBC, electrolytes, renal function, liver function and thyroid function; beta-hCG to rule out pregnancy before prescribing drugs contraindicated in pregnant patients
 - metabolic parameters including weight, waist circumference, lipid profile and fasting blood sugars at the time of initiation of any treatment
 - for patients with any unusual symptoms, neurologic signs or symptoms, or a first manic or hypomanic episode after age 40, include structural imaging of the brain such as CAT or MRI scan

Therapeutic Choices

Bipolar disorder is a complex illness that can only be partly managed by a sole physician; a team approach is ideal to ensure the patient receives education, ongoing monitoring and support. The CANMAT/ISBD treatment guidelines^[6] provide details for the comprehensive treatment of bipolar disorder and are the basis for the recommendations that follow. Additionally, the guidelines list key resources such as treatment manuals and web sites.

Nonpharmacologic Choices

- **Psychoeducation** is a recommended intervention consisting of information about the illness as well as training in coping strategies and recognition of episodes early in their genesis.^{[7][8]} Structured group psychoeducational programs have been shown to substantially reduce the risk of relapse of manic, mixed and depressive episodes; group or family formats appear to confer most benefit.^[9] *Structured Group Psychotherapy for Bipolar Disorder* (see [Suggested Readings](#)) is a recommended resource. Available in both English and French, it provides explicit advice for effective psychoeducation over the course of 6 treatment sessions.
- A patient self-help book by Bauer et al uses this model and is highly recommended for use with patients (see [Suggested Readings](#)).
- Evidence also supports **cognitive behavioural therapy** and **family therapy**, for relapse risk reduction.^[9]
- Patients who are well-informed and engaged in the management of their mood disorder may enjoy a better quality of life. Many online self-help resources are available for patients, their families and health-care providers:
 - [the Patient and Family Guide to the CANMAT and ISBD Guidelines on the Management of Bipolar Disorder](#)
 - the [patient resources section](#) of the International Society for Bipolar Disorders website provides basic information about the condition and strategies for developing coping skills; the website also includes excellent resources for clinicians
 - the Collaborative REsearch Team to study psychosocial issues in Bipolar Disorder ([CREST.BD](#)) offers a self-management toolkit for patients and information about the psychosocial issues of bipolar disorder
 - [eMoods](#) provides graphs of mood over time through its mood tracking app
- To prevent an incipient manic or depressive episode from becoming a flagrant episode, conduct a “relapse drill” that trains patients to recognize their unique warning symptoms and to respond to these symptoms with a specific treatment plan.^[10] Some of these responses might include:
 - changes in medication strategies

- rapid contact with the treating physician
- additional steps to regulate sleep and other behaviours
- Key principles for addressing cognitive impairment include:^[11]
 - meticulous attention to treating bipolar disorder to full remission
 - treating concomitant disorders such as substance use disorders, anxiety disorders and ADHD
 - possible discontinuation of medications that may impair cognition (e.g., medical marijuana, benzodiazepines, drugs with high anticholinergic burden, including antiemetics, antihistamines, antimuscarinics and some antidepressants and antipsychotics)
 - rigorous management of major medical comorbidities, particularly thyroid disease, which often has subclinical manifestations
 - promotion of wellness routines known to improve cognition, such as careful sleep regulation and regular exercise

Pharmacologic Choices

Pharmacologic therapy varies according to the type and stage of the episode being treated. Since treatment for this condition is lifelong, engaging the patient in collaborative decision-making is critical. Except in situations where there is a severe episode or medical emergency, discuss the pros and cons of 2 or 3 pharmacologic options to inform and support the patient's decision. A newer approach recommended in the 2018 CANMAT guidelines involves balancing treatment recommendations for an acute episode with knowledge of how that compound works for other phases of the disorder, since medication treatment is often long-term.^[6] For instance, in choosing between 2 medications with similar evidence of efficacy for acute mania, where 1 has better evidence for efficacy in the maintenance phase, consider choosing that medication first.

Manic Episodes

Moderate to severe mania is often treated in hospital; mild mania, which by psychiatric definition is distinct from hypomania, may be treated on an outpatient basis. The first step in treating mania is to assess for risk of aggressive behaviour or violence to others, suicide, degree of insight and the ability to adhere to treatment (see [Figure 1](#)).^[6] If the patient is taking an antidepressant, it should be immediately discontinued. Complicating conditions, particularly substance use disorders, will need attention.

Specific medication strategies for mania depend on whether the patient is already on maintenance therapy and is experiencing a breakthrough episode or whether the individual is unmedicated. If the patient is already taking a first-line agent such as **lithium**, **divalproex** or a **second-generation (“atypical”) antipsychotic**, dosage adjustment may be sufficient after checking blood levels where appropriate. In moderate-to-severe manic episodes, addition of another medication is usually necessary.

In previously unmedicated patients, initiate treatment with a first-line agent (see [Table 3](#), [Table 6](#)). Lithium remains “first among equals,” since it is advantageous in the overall management of bipolar disorder with additional anti-suicidal properties and may also possibly minimize risk of dementia.^{[12][13][14][15][16]}

When the episode is particularly severe, initiate treatment with a 2-drug combination such as lithium or divalproex plus a second-generation antipsychotic. Continue treatment on any particular regimen for 2 weeks at therapeutic doses before assessing whether a change is necessary. Various other treatment options, including ECT (effective for both mania and depression), are shown in [Figure 1](#).

Medication selection must balance the side effect profile with key clinical characteristics, particularly prior response and patient preferences. Additionally, some safety considerations differ by medication:

- Lithium: caution with renal disease
- Divalproex: caution in patients of reproductive age due to effects on menstrual irregularities and teratogenicity during pregnancy
- Carbamazepine: caution due to drug-drug interactions and induction of Stevens-Johnson syndrome
- Antipsychotics: caution due to adverse effects on metabolic parameters

For manic episodes, the optimal duration of the second-generation antipsychotic in a 2-drug combination has not been determined, although a randomized clinical trial suggests that treatment for 6 months yields the most favourable risk-benefit ratio; decrease in relapse rate was similar to that achieved with treatment for 12 months, and significant weight gain due to metabolic side effects did not occur within 6 months.^[17]

Table 3: Pharmacologic Management of Acute Mania in Bipolar Disorder in Order of Preference^[6]

Place in Therapy	Drugs ^{[a][b]}	Comments
First-line	lithium	While high blood lithium level produce more response, it involves more risk. Consider moderate blood levels (0.8–1.0 mmol/L) and add other anti-manic agents if necessary. Because of overall value, consider lithium first, particularly in younger adults.
	quetiapine	Rapid dose titration is generally safe, but monitor for hypotension.
	divalproex	Loading dose is an option (20 mg/kg); aim for blood levels high in therapeutic range (350–800 mcmol/L).
	asenapine	Sublingual tablet; important not to eat or drink for 10 min after administration.

Place in Therapy	Drugs ^{[a][b]}	Comments
	aripiprazole	Limited benefit for depressed phase, less desirable in someone with a history of frequent depressive relapses.
	paliperidone >6 mg	Available as extended-release tablet and long-acting IM injection.
	risperidone	Monitor regularly for orthostatic hypotension and extrapyramidal side effects, e.g., parkinsonism, akathisia.
	cariprazine	Effective for both mania and depression; higher doses used for mania and lower doses used for depression.
	lithium or divalproex <i>plus</i> quetiapine	Allows lower dosing of quetiapine than in monotherapy; useful for more severe episodes; start quetiapine first × 1–3 days.
	lithium or divalproex <i>plus</i> aripiprazole	Better tolerated than most other combination treatments.
	lithium or divalproex <i>plus</i> risperidone	Often can taper risperidone after 6 months of euthymia. ^[17] Allows lower dosing of risperidone than in monotherapy; useful for more severe episodes; start risperidone first × 1–3 days.
	lithium or divalproex <i>plus</i> asenapine	Relatively well tolerated.
Second-line	Try several first-line treatments first before going to second-line treatments.	
	olanzapine	Slightly more effective than other SGAs, but far more long-term side effects.

Place in Therapy	Drugs ^{[a][b]}	Comments
	carbamazepine (including extended-release)	Watch for hepatic induction, which lowers blood levels of many medications, including key SGAs and some antidepressants.
	lithium or divalproex <i>plus</i> olanzapine	Good for severe episodes, but highest side effect burden in the long term.
	lithium <i>plus</i> divalproex	Less beneficial than combinations of a mood stabilizer plus SGA.
	ziprasidone	Limited evidence in maintenance, not effective for depression, cardiac monitoring at higher doses.
	haloperidol	Well tolerated and highly efficacious acutely, but promotes depression relapse and tardive dyskinesia concerns in the long term.
	ECT	Appropriate and effective for severe mania, and in medically ill.
Third-line	Many potential agents, to be used by specialists, are very useful in rapid-cycling patients.	
Not recommended	Monotherapy with allopurinol, eslicarbazepine/licarbazepine, gabapentin, lamotrigine, omega-3 fatty acids, topiramate, valnoctamide, zonisamide Combination therapy with carbamazepine plus risperidone or olanzapine	Use of some of these agents as adjuncts may be appropriate, especially if initiated prior to mania for other symptoms, e.g., using gabapentin for anxiety, topiramate for alcohol craving. To be implemented by a specialist.

^[a] Pharmacologic treatment of hypomanic episodes in bipolar II disorder is similar.

^[b] Treatment trials typically run 2 wk at therapeutic doses. Adjunctive benzodiazepine use is common, e.g., clonazepam 1–2 mg BID. If patient is not on first-line agent, start first-line agent monotherapy; if more serious, consider dual therapy. If on maintenance therapy, optimize dosing prior to adding new medication. Adapted with permission from Yatham LN, Kennedy SH, Parikh SV et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) 2018 guidelines for the management of patients with bipolar disorder. *Bipolar Disord* 2018;20(2):97-170.

Abbreviations: ECT = electroconvulsive therapy; SGA = second generation antipsychotic

Depressive Episodes

As with acute mania, first assess the patient for basic safety issues, including potential suicidality, comorbid medical problems or substance use. Next, the strategy depends on whether the patient is on medication and has had a breakthrough episode of major depression, or whether they are medication free (see [Figure 2](#)).^[6] In an unmedicated patient, therapy may begin with any of the first-line treatments: **lithium**, **lamotrigine**, **lurasidone** or **quetiapine** as monotherapies (see [Table 4](#), [Table 7](#)). Lamotrigine is the best tolerated option, although it may take the longest to work and may be slightly less efficacious.

Another first-line option is **cariprazine**, an atypical antipsychotic approved in Canada in 2022 for the treatment of both bipolar depression and mania.^{[19][20]}

Additional first-line adjunctive strategies include lurasidone with either lithium or divalproex; or addition of lamotrigine to another first-line agent. If the depression is severe, therapy can be initiated with 2 first-line agents, such as lithium plus quetiapine. Efficacy of lurasidone as monotherapy or in combination with lithium or valproate for adults with bipolar depression has been demonstrated in RCTs.^{[21][22]} Anxiety, EPS, headache, insomnia and nausea were the most common adverse effects reported in the RCTs and in a 24-week open-label extension trial.^[23] These studies also demonstrated that lurasidone has a low risk of weight gain and metabolic disturbances.

Bipolar depression is difficult to treat; steps to follow if there is an inadequate response after 2–4 weeks of therapy are outlined in [Figure 2](#). Two key emerging observations are that specific psychosocial strategies are potentially valuable for bipolar depression and that there are a number of second-line treatments. Given the many potential complications of adding adjunctive **antidepressants**, as well as their limited efficacy,^[24] this strategy is clearly second-line. **SSRIs** and **bupropion** are options that may be effective and have a limited ability to provoke a switch into mania.^[25] Avoid antidepressants in the presence of a current mixed episode, a history of rapid cycling, or a history of antidepressant-induced mania.^[25]

Serious bipolar depression merits consideration of ECT or the use of agents with heavier side-effect concerns, such as the combination of olanzapine plus fluoxetine. Combining 2 traditional **mood stabilizers** such as **lithium** plus **divalproex** may also be of some value; alternatively, using divalproex as monotherapy may be considered. Intensive psychosocial interventions for acute bipolar depression also have some efficacy, compared with brief psychosocial intervention, while usual bipolar medications are maintained.^[26]

While not approved, there is strong evidence to support IV **ketamine** for the treatment of unipolar depression and modest evidence for bipolar depression; frequency of administration and duration of treatment remain unclear.^[27] The intranasal enantiomer **esketamine** is approved for treatment of unipolar

depression, and preliminary evidence suggests value for bipolar depression.^[28] Additionally, **lumapterone** has been approved for schizophrenia internationally and has been studied in several clinical trials for bipolar depression; these studies support its antidepressant effect and are undergoing review for approval for bipolar depression by the US FDA.

Table 4: Pharmacologic Management of Acute Depression in Bipolar Disorder in Order of Preference^[6]

Place in Therapy	Drugs ^[a]	Comments
First-line	quetiapine (including extended-release)	Monotherapy studies showed doses of 300 or 600 mg/day to be equally effective; lower dose preferred if patient is medication-naïve.
	lithium or divalproex <i>plus</i> lurasidone ^[22]	Lurasidone must be taken with food—at least 350 calories. Initiate lurasidone at 20 mg/day; titrate to lowest effective dose to minimize adverse effects.
	lithium	Aim for blood levels near the high end of therapeutic range as tolerated, i.e., 1–1.2 mmol/L (0.5–0.8 mmol/L in elderly patients). Measure serum concentration 9–13 h post-dose. May be useful for those with high risk of suicide or self harm. ^[18]
	lamotrigine	Slow titration schedule slows response but mitigates risk of adverse effects including rash; useful for milder episodes.
	lurasidone ^{[21][23]}	Must be taken with food—at least 350 calories. Initiate at 20 mg/day; titrate to lowest effective dose to minimize adverse effects.
	adjunctive lamotrigine	Slow titration schedule slows response but mitigates risk of adverse effects including rash.

Place in Therapy	Drugs ^[a]	Comments
	cariprazine	The CANMAT Bipolar Guideline Committee (of which the author is a member) recommends cariprazine as a first-line option based on clinical data and experience established since the publication of 2018 guidelines. ^{[19][20]}
Second-line	divalproex adjunctive SSRI or bupropion ECT olanzapine <i>plus</i> fluoxetine^[b]	Try several first-line treatments before going to second-line options.
Third-line	many choices exist (almost always as adjunctive treatments), including selected atypical antipsychotics, SNRI or MAOI antidepressants, IV ketamine , stimulants, light therapy , rTMS, and others	Third-line choices best initiated by mood disorder specialists. ECT may be initial treatment in particularly severe depression or in medically ill patients.
Not recommended	monotherapy with antidepressants, gabapentin, aripiprazole or ziprasidone adjunctive ziprasidone or levetiracetam lamotrigine plus folic acid mifepristone	

^[a] Treatment trials typically run a minimum of 2–4 wk at therapeutic doses. Response is often slower than in unipolar depression. Avoid antidepressants if possible, or shorten duration of antidepressant treatment to prevent switch into mania. If the depression is severe, therapy can be initiated with 2 first-line agents, such as lithium plus quetiapine.

^[b] Adverse effects, especially those affecting metabolic profile, limit use

Adapted with permission from Yatham LN, Kennedy SH, Parikh SV et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) 2018 guidelines for the management of patients with bipolar disorder. *Bipolar Disord* 2018;20(2):97-170.

Abbreviations: ECT = electroconvulsive therapy; MAOI = monoamine oxidase inhibitor; rTMS = repetitive transmagnetic stimulation; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; TCA = tricyclic antidepressant

Maintenance Therapy for Bipolar Disorder

When patients recover from an acute episode and remain well (with no or minimum symptoms) for at least 2 months, they are defined as being in remission and hence enter the maintenance phase of the illness. Without pharmacologic treatment, such individuals have an approximately 70% chance of experiencing another episode over the next year and an approximately 95% chance of having a recurrence within 5 years. Since nonadherence to treatment is the principal risk factor for recurrence, adopting a collaborative decision-making stance and inquiring about adherence and attitudes to treatment in a non-judgmental way is essential.

To prevent relapse, interventions include psychosocial strategies such as psychoeducation, cognitive-behavioural therapy, family therapy, and interpersonal and social rhythm therapy. Key characteristics of effective psychosocial therapies that improve medication adherence include education about the disorder (including the likelihood of relapse), training in self-monitoring, education about how to manage side effects, development of strategies to manage stressors and attention to the patient’s own belief system and attitude toward the illness.

Give patients periodic occasions to discuss the impact of the illness on their life and their beliefs about potential effects of long-term medication use. To facilitate longer-term adherence, repeat a brief version of psychoeducation periodically, e.g., annually.

[Table 5](#) and [Table 8](#) list a number of effective options for relapse prevention. [Figure 3](#) outlines a step-wise approach to maintenance therapy.

Given the efficacy of **cariprazine** for mania, depression and mixed episodes, it is anticipated to be proven effective for the maintenance phase as well; studies evaluating this have not been completed.

Optimization of cognition is an emerging target for treatment, as a substantial proportion of patients have problems in various cognitive domains, even in remission. No medications have been established as efficacious for these deficits, although stimulants used for ADHD are often used in clinical practice. For more information on non-pharmacologic approaches, see [Nonpharmacologic Choices](#).

Table 5: Maintenance Treatment Regimens for Bipolar Disorder^{[a][6]}

Place in Therapy	Drugs	Comments
First-line	lithium	Blood levels can be individualized—some patients have efficacy at lower end of normal therapeutic range (0.5–0.8 mmol/L); levels lower than for acute

Place in Therapy	Drugs	Comments
		episode. Effective in preventing manic and depressive relapses.
	quetiapine	Effective in preventing both manic and depressive relapses.
	divalproex	Moderate evidence supports effectiveness in both manic and depressive relapses.
	lamotrigine	Effective primarily for prophylaxis against depressive relapses.
	asenapine	Effective for both manic and depressive relapses.
	lithium or divalproex <i>plus</i> quetiapine	Effective for both manic and depressive relapses.
	lithium or divalproex <i>plus</i> aripiprazole	Effective primarily for prevention of manic relapse.
	aripiprazole	Effective for prevention of manic relapse.
	aripiprazole OM monotherapy	This depot injection is effective for prevention of manic relapse.
Second-line	Try several first-line treatments before going to second-line treatments.	
	olanzapine	Effective for preventing both manic and depressive relapses, but with high side effect burden.
	risperidone LAI	Long-acting injection effective for preventing manic relapses.

Place in Therapy	Drugs	Comments
	adjunctive risperidone LAI	Long-acting injection effective for preventing manic relapses.
	carbamazepine	Moderate evidence for prevention of both manic and depressive relapses, with significant side effect concerns.
	paliperidone (>6 mg)	Primarily for prevention of manic relapse.
	lithium or divalproex <i>plus</i> lurasidone	Limited evidence suggests effective for preventing depressive relapse, and very limited evidence suggests effective for preventing manic relapse.
	lithium or divalproex <i>plus</i> ziprasidone	Primarily prevents manic relapse.
Third-line	aripiprazole <i>plus</i> lamotrigine adjunctive clozapine adjunctive gabapentin olanzapine <i>plus</i> fluoxetine	Third-line choices best reserved for mood disorder specialists.
Not recommended	perphenazine TCAs	Patients with bipolar II disorder rarely may benefit from monotherapy with antidepressants.

^[a] While the goal is ultimately complete relapse prevention, in individuals with severe illness, reduction in the frequency, duration and intensity of relapse may be more realistic. Psychosocial interventions are critical to reduce relapse.

Adapted with permission from Yatham LN, Kennedy SH, Parikh SV et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) 2018 guidelines for the management of patients with bipolar disorder. *Bipolar Disord* 2018;20(2):97-170.

Abbreviations: ECT = electroconvulsive therapy; LAI = long-acting injection; OM = once monthly; RCT = randomized controlled trial; TCA = tricyclic antidepressant

Bipolar Disorder in Children and Adolescents

Bipolar disorder typically emerges in adolescence, most often as depression.^[29] An initial depressive episode is impossible to distinguish from unipolar depression. The existence of first-degree relatives with bipolar disorder in an adolescent presenting with an initial depression is suggestive of an eventual bipolar diagnosis, but treatment is usually initiated on the assumption that the teenager has unipolar depression. Other key predictors of an eventual shift to a bipolar diagnosis include the presence of particularly severe or psychotic symptoms in a teen who is unmistakably depressed (as opposed to someone who likely has schizophrenia). The diagnostic criteria of adult bipolar disorder should be applied to correctly diagnose bipolar disorder in children, with a new emphasis on understanding that irritability is present in a variety of childhood psychiatric disorders, and is not particularly suggestive of mania or bipolar disorder in children.

Both mania and depression have been documented in preteens, therefore bipolar disorder may be reliably diagnosed by a specialist at a very early age. Attention-deficit hyperactivity disorder (ADHD) presents a major differential challenge, with the key distinguishing feature being the episodic nature (weeks or months) of the bipolar disorder, as opposed to ADHD, where symptoms are present on a continuous daily basis. When comorbid bipolar disorder and ADHD are simultaneously diagnosed, treat the patient for bipolar disorder first and offer ADHD treatment when the bipolar disorder has stabilized.^[30] Treatment options for bipolar disorder are limited by a relative lack of clinical trials in children and adolescents, but for treating mania, first-line recommendations include **lithium**, **risperidone**, **aripiprazole**, **asenapine** and **quetiapine**. For pediatric bipolar depression, modest evidence supports a first-line recommendation of **lurasidone**, and second-line treatments include lithium and **lamotrigine**.^[6] Given the various diagnostic and medication complexities associated with children, refer to a child psychiatrist for treatment.

Bipolar Disorder in the Elderly

When bipolar disorder begins early in life, it is a lifelong illness that, if untreated, continues to manifest with more frequent episodes and shorter periods of recovery over the decades. However, the illness does not typically progress to an extremely rapid-cycling form of illness, and elderly bipolar patients usually will have major periods of euthymia. Cognitive dysfunction is a significant feature for many elderly bipolar individuals and needs regular assessment and management. Due to high rates of comorbid medical disorders and resultant polypharmacy, these individuals should have periodic physical and neurologic exams, careful lab monitoring, and may need neuroimaging.

A small number of individuals experience a first onset of bipolar illness in their 60s or later; symptoms are typical of adult mania but irritability may be prominent.^[31] Often these individuals have a history of comorbid neurologic problems and an elevated all-cause mortality rate.

Clinical trials are relatively rare in the elderly, so most treatment parameters are extrapolated from the adult literature.^[32] For mania, first-line treatments include **lithium** or **divalproex**, with **quetiapine** recommended as the only second-line agent. Third-line treatments include a number of other atypical antipsychotics, with **clozapine** and **ECT** as options for refractory episodes. For bipolar depression in older individuals, options include **lurasidone** and quetiapine initially, with lithium and **lamotrigine** as the second-line recommendations. ECT should be considered for refractory bipolar depression or for situations of high suicide risk or inability to maintain adequate food and fluid intake. Maintenance treatment will depend on which medications were helpful for the acute episode and remain safe and tolerable.

All second-generation antipsychotics carry warnings about potential elevated stroke risk and mortality in the elderly, but the significant morbidity and mortality associated with bipolar disorder in this population compels consideration of these agents for treatment.

Choices during Pregnancy and Breastfeeding

Since bipolar disorder often starts early in life, many patients will be living with the disorder at the time they are considering having children. During family planning, potential risks to the developing fetus should be considered when selecting between pharmacologic options.^[6] Key management principles include careful risk assessment for each patient, namely the risk of pregnancy destabilizing the illness and the possibility of severe episodes resulting in the death of either the patient or child or both. Untreated patients with bipolar disorder are particularly susceptible to postpartum depression, including depression with psychotic features that may result in harm to the child. Despite such cautions, most patients with bipolar disorder are able to manage pregnancy successfully. Collaborative management of the pregnancy by a psychiatrist, an obstetrician and a family physician is recommended, including consultation with an agency specializing in information on medications during and following pregnancy, such as [MothertoBaby](#). If medication is to be discontinued prior to conception, it should be tapered with medical supervision.^[33]

Creation of a “pregnancy contract” is recommended: in consultation with the doctor, the patient creates a 1-page document that summarizes their typical symptoms in mania and in depression and lists their preferred treatment options for each phase of illness. That document is then shared among the treatment team and the patient’s family, and serves as a nonbinding but helpful advance directive should a new episode of illness arise.

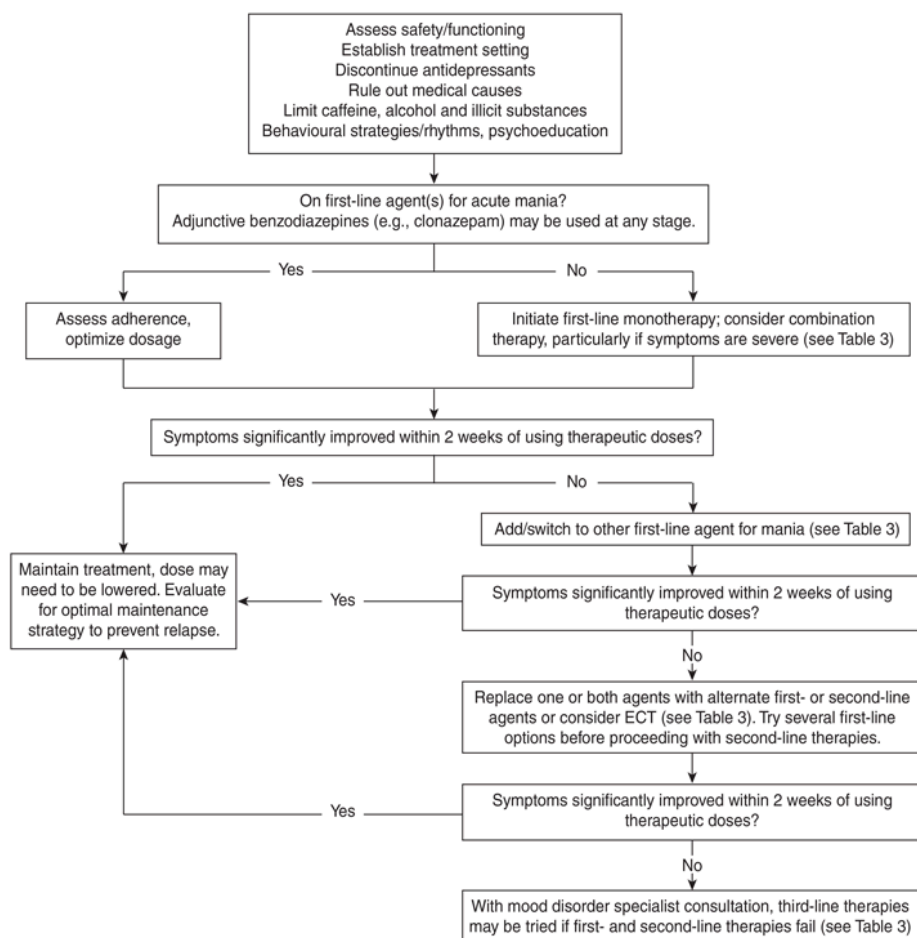
Overall, the management of medications during pregnancy is complex and beyond the scope of this chapter and should be guided by the detailed advice in the CANMAT bipolar guidelines.^[6] A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- A key feature of bipolar disorder is recurrent nonadherence to medication; including the patient in shared decision-making, together with psychoeducation, promotes a strong therapeutic alliance and enhances medication adherence.
- Despite the profusion of newer treatments, **lithium** is still considered the most important medication in treating bipolar disorder, and should be offered early in the course of illness.
- Patients taking lithium need to maintain their usual salt and caffeine intake and monitor fluid intake and output, making adjustments in the event of unexpected losses due to vomiting or diarrhea.
- During acute manic episodes, patients may exhibit increased tolerance to lithium. Therefore, higher doses may be needed during mania, followed by dose reduction after mania resolves.
- Advise patients taking antipsychotics about antipsychotic-associated body temperature dysregulation and strategies to prevent heat stroke, e.g., hydration, sun protection.
- For lithium-associated cognitive impairment, check lithium level and thyroid function. Lowering the dose or using a slow-release formulation may improve cognitive function. Despite earlier literature recommendations of higher lithium levels, low to medium levels (0.5–0.8 mmol/L) are increasingly suggested in maintenance, with the switch or addition of another maintenance medication strategy preferred over pushing the lithium blood level higher.
- Patients who experience tremor while taking lithium may benefit from elimination of dietary caffeine, lithium dose reduction or the addition of a beta-blocker such as **propranolol** or **atenolol**.
- Patients who experience diarrhea while taking slow-release lithium preparations may fare better with immediate-release formulations.^[34]

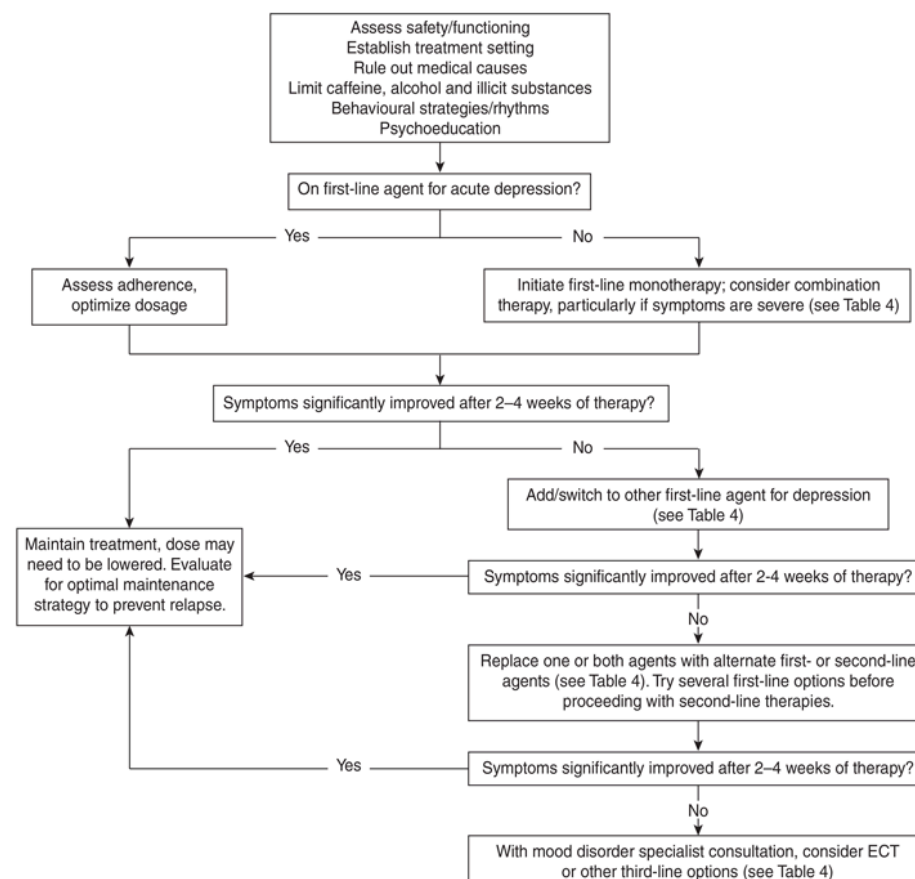
Algorithms

Figure 1: Management of Acute Mania in Bipolar Disorder



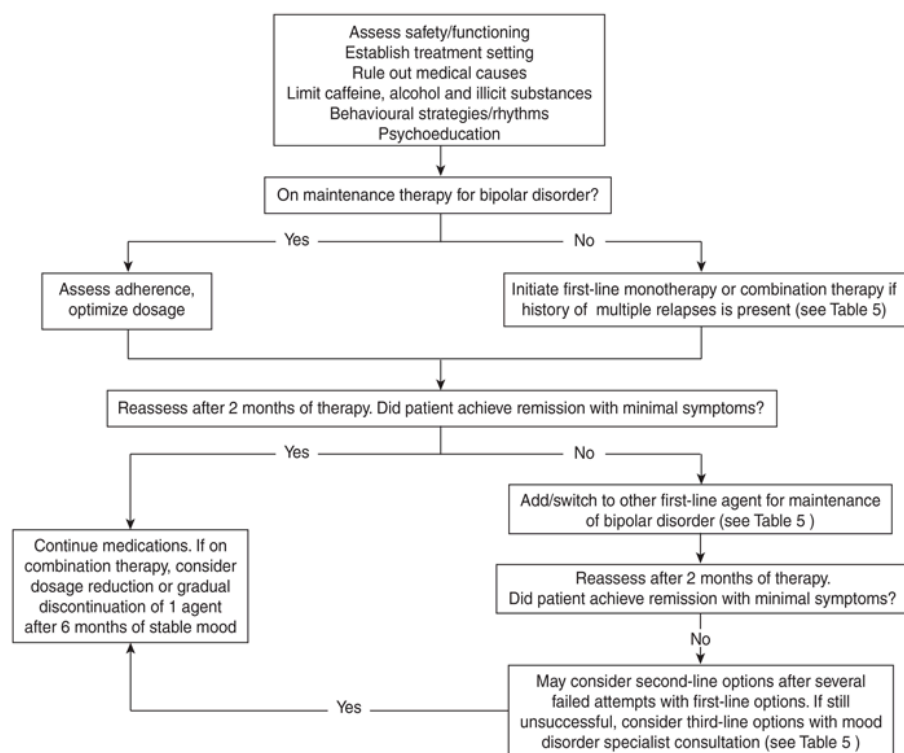
Abbreviations: ECT = electroconvulsive therapy

Figure 2: Management of Depressive Episodes in Bipolar Disorder



Abbreviations: ECT = electroconvulsive therapy

Figure 3: Management of Maintenance Therapy for Bipolar Disorder



Drug Tables

Table 6: First-line Drug Therapy for Mania in Bipolar Disorder

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, second-generation				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
aripiprazole Abilify, generics \$30–60	Initial: 15 mg/day PO in monotherapy or 10– 15 mg/day PO if combined with lithium or divalproex Dose can be increased after 2 wk Maximum: 30 mg/day	Akathisia, dizziness, orthostatic hypotension, headache, GI complaints, tremor, sedation. Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4 such as phenytoin, rifampin) can reduce aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially.	First-line as monotherapy or in combination with lithium or divalproex.
asenapine Saphris \$90–120	Initial: 5 mg BID SL Dose can be slowly increased to a maximum of 10 mg BID SL depending on response and tolerability Instruct patients not to eat or drink for 10 min after placing tablet under tongue	Sedation, orthostasis, hypotension, dizziness, weight gain, EPS . Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Decreased asenapine clearance with fluvoxamine.	First-line as monotherapy or in combination with lithium or divalproex.
cariprazine Vraylar ~\$160	Initial: 1.5 mg once daily Dose can be increased by 1.5 mg increments to a	Akathisia, EPS , dyspepsia, vomiting, somnolence, restlessness. Less commonly	Extensively metabolized by CYP3A4; contraindicated in combination with moderate/strong CYP3A4	First-line as monotherapy or in combination with lithium or divalproex.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	maximum of 6 mg once daily	orthostatic hypotension. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	inhibitors/inducers and for at least 2 wk after discontinuation (inhibitors).	Effective for both mania and depression.
paliperidone  extended-release Invega \$120–370	Initial: 3–6 mg/day PO Dose can be titrated up by increments of 3 mg/day every 3–5 days Maximum: 12 mg/day	Insomnia, headaches, weight gain, orthostasis, rhinitis, anxiety, dose-related hyperprolactinemia and EPS . Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Minimal risk of drug interactions. Carbamazepine may decrease paliperidone concentrations.	Considered first-line at doses >6 mg/day. On the NIOSH list of non-antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and disposing of this drug. ^[b]
quetiapine Seroquel, generics < \$30	Day 1: 50 mg BID PO Day 2: 100 mg BID PO Day 3: 150 mg BID PO Day 4: 200 mg BID PO	Weight gain, sedation, orthostasis, metabolic disturbances. Monitor periodically for movement disorders, including akathisia.	Ketoconazole dramatically increases quetiapine levels. Watch for hypotension with antihypertensives.	First-line as monotherapy or in combination with lithium or divalproex.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	Day 5 (if necessary): 300 mg BID PO Day 6 (if necessary): 400 mg BID PO Usual dose: 200–400 mg BID	Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		
risperidone  Risperdal , Apo-Risperidone, other generics \$30–60	Initial: 2–3 mg/day PO Dose can be titrated up by increments of 1 mg/day every 24 h Usual dose: 3–6 mg/day in a single daily dose or 2 divided doses	Weight gain, orthostasis, dose-related EPS , metabolic disturbances. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Avoid combining with carbamazepine.	First-line as monotherapy or in combination with lithium or divalproex. On the NIOSH list of non-antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and disposing of this drug. ^[b]
Drug Class: Mood Stabilizers				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
divalproex sodium Epival , Apo-Divalproex, Mylan-Divalproex , other generics < \$30	Initial: 250 mg TID PO Dose can be titrated up by increments of 250–500 mg/day every 1–3 days depending on tolerability Usual dose: 750–2000 mg/day PO, adjusted according to serum levels (target range 350–800 mcml/L)	Weight gain, thrombocytopenia, polycystic ovary syndrome, teratogenic, hyperammonemia (rare), hepatotoxicity (rare).	Avoid combining with carbamazepine. Use caution if combining with lamotrigine.	First-line as monotherapy or in combination with an SGA. On the NIOSH list of non-antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and disposing of this drug. ^[b]

lithium  carbonate immediate-release Carbolith, Lithane, Apo-Lithium Carbonate, other generics < \$30	Initial: 300 mg BID PO (150 mg BID PO in the elderly) Dose can be titrated up by increments of 300–600 mg/day every 5–7 days, depending on tolerability and response Usual: 900–2100 mg/day PO, guided by serum concentrations (target 0.8–1 mmol/L; 0.5–0.8 mmol/L in elderly patients) Measure serum concentrations 9–13 h post-dose	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment. Monitor thyroid and renal function at least every 6 months. Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term treatment with these agents is required. Avoid large changes in salt or coffee intake.	First-line as monotherapy or in combination with an SGA. Stop lithium temporarily during acute intermittent illnesses causing fluid and electrolyte losses.
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Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	Once effective dose is reached, single daily dose is preferred			

lithium  carbonate controlled-release Lithmax \$30–60	Initial: 300 mg BID PO (150 mg BID PO in the elderly) Dose can be titrated up by increments of 300–600 mg/day every 5–7 days, depending on tolerability and response Usual: 900–2100 mg/day PO, guided by serum concentrations (target 0.8–1 mmol/L; 0.5–0.8 mmol/L in elderly patients) Measure serum concentrations 9–13 h post-dose Once effective dose is reached, single daily dose is preferred	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment. Monitor thyroid and renal function at least every 6 months. Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term treatment with these agents is required. Avoid large changes in salt or coffee intake.	First-line as monotherapy or in combination with an SGA. Stop lithium temporarily during acute intermittent illnesses causing fluid and electrolyte losses.
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valproic acid Depakene , Apo-Valproic, other generics < \$30	Initial: 250 mg TID PO Dose can be titrated up by increments of 250–500 mg/day every 1–3 days depending on tolerability	Weight gain, thrombocytopenia, polycystic ovary syndrome, teratogenic, hyperammonemia (rare),	Avoid combining with carbamazepine. Use caution if combining with lamotrigine.	First-line as monotherapy or in combination with an SGA. On the NIOSH list of non-
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Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	Usual dose: 750–2000 mg/day PO, adjusted according to serum levels (target range 350–800 mcmol/L)	hepatotoxicity (rare).		antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and disposing of this drug. ^[b]

^[a] Cost of 30-day supply of mean usual dose (unless specific cost range provided); includes drug cost only.

^[b] For more information, see www.cdc.gov/niosh/docs/2016-161/.

⚠ Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; EPS = extrapyramidal symptoms; GI = gastrointestinal; NIOSH = National Institute for Occupational Safety and Health; NSAID = nonsteroidal anti-inflammatory; SGA = second-generation antipsychotic

Table 7: First-line Drug Therapy for Depression in Bipolar Disorder^[a]

Drug/Cost ^[b]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, second-generation				
lurasidone ⚠ Latuda, generics	Initial: 20 mg/day PO with food (at least 350 calories)	Akathisia, anxiety, EPS , insomnia, nausea/vomiting, somnolence. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	May potentiate antihypertensive drug effects; do not use in combination with strong inhibitors of CYP3A4 such as ketoconazole; do not exceed doses of 40 mg daily PO in combination with other inhibitors of CYP3A4 such as diltiazem, clarithromycin, erythromycin, prednisone, grapefruit, grapefruit juice	First-line as monotherapy or in combination with lithium or divalproex.
\$25–50	Usual range: 20–60 mg/day Maximum: 120 mg/day	Monitor patient for lurasidone-associated NMS		

Drug/Cost ^[b]	Dosage	Adverse Effects	Drug Interactions	Comments
		and discontinue if NMS is suspected.	combination with strong inducers of CYP3A4 such as carbamazepine, phenytoin (decreased lurasidone levels).	
cariprazine Vraylar	Initial: 1.5 mg once daily	Akathisia, nausea, restlessness and EPS . Less commonly orthostatic hypotension. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Extensively metabolized by CYP3A4; contraindicated in combination with moderate/strong CYP3A4 inhibitors/inducers and for at least 2 wk after discontinuation (inhibitors).	First-line option. Effective for both mania and depression.
~\$160	Dose can be increased to 3 mg once daily on day 15, if needed; maximum 3 mg/day			
quetiapine Seroquel, Seroquel XR, generics	Day 1: 50 mg/day PO Day 2: 100 mg/day PO Day 3: 200 mg/day PO Day 4: 300 mg/day PO Usual dose: 300 mg/day PO Maximum: 600 mg/day PO	Weight gain, sedation, orthostasis, metabolic disturbances. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Ketoconazole dramatically increases quetiapine levels. Monitor for hypotension with antihypertensives.	
< \$25				
Drug Class: Mood Stabilizers				

Drug/Cost ^[b]	Dosage	Adverse Effects	Drug Interactions	Comments
lamotrigine Lamictal , generics < \$25	Initial: 25 mg once daily PO Increase to 25 mg BID PO after 2 wk, then by 25–50 mg/day increments every 1–2 wk Usual dose: 100–300 mg/day PO	Rash (common), severe rash (rare). Slow titration reduces risk of severe dermatological reactions.	Divalproex doubles blood level of lamotrigine; lower initial and target doses are recommended.	
lithium carbonate immediate-release Carbolith, Lithane, Apo-Lithium Carbonate, other generics < \$25	Initial: 300 mg BID PO (150 mg BID PO in the elderly) Dose can be titrated up by increments of 300–600 mg/day every 5–7 days, depending on tolerability and response Usual: 600–1800 mg/day PO, guided by serum concentrations (target 1–1.2 mmol/L; 0.5–0.8 mmol/L in elderly patients) Measure serum concentrations 9–13 h post-dose Once effective dose is reached, single daily dose is preferred	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment. Monitor thyroid and renal function at least every 6 months. Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term treatment with these agents is required. Avoid large changes in salt or coffee intake.	Stop lithium temporarily during acute intermittent illnesses causing fluid and electrolyte losses.

Drug/Cost ^[b]	Dosage	Adverse Effects	Drug Interactions	Comments
lithium carbonate controlled-release Lithmax \$25–50	Initial: 300 mg BID PO (150 mg BID PO in the elderly) Dose can be titrated up by increments of 300–600 mg/day every 5–7 days, depending on tolerability and response Usual: 600–1800 mg/day PO, guided by serum concentrations (target 1–1.2 mmol/L; 0.5–0.8 mmol/L in elderly patients) Measure serum concentrations 9–13 h post-dose Once effective dose is reached, single daily dose is preferred	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment. Monitor thyroid and renal function at least every 6 months. Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term treatment with these agents is required. Avoid large changes in salt or coffee intake.	Stop lithium temporarily during acute intermittent illnesses causing fluid and electrolyte losses.

^[a] Detailed information on antidepressants can be found in [Depression](#).
^[b] Cost of 30-day supply for mean dose; includes drug cost only.
 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).
Abbreviations: ARB = angiotensin receptor blocker; EPS = extrapyramidal syndrome; NMS = neuroleptic malignant syndrome

Table 8: First-line Maintenance Therapy for Bipolar Disorder

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, second-generation				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
aripiprazole Abilify, generics \$30–60	15 mg/day PO Maximum: 30 mg/day	Akathisia, dizziness, GI complaints, orthostatic hypotension, headache, sedation, tremor. Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4 such as phenytoin, rifampin) can decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially.	First-line as monotherapy or in combination with lithium or divalproex.
aripiprazole prolonged- release injection Abilify Maintena \$485/vial	400 mg once monthly by deep IM injection in the gluteal or deltoid muscle Dose can be reduced to 300 mg once monthly if adverse reactions occur	Akathisia, dizziness, GI complaints, orthostatic hypotension, headache, sedation, tremor. Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4 such as phenytoin, rifampin) can decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially.	Continue oral aripiprazole during 14 consecutive days after the first injection to maintain therapeutic concentrations.
asenapine Saphris \$90–120	5 mg BID SL Maximum: 10 mg BID SL Instruct patients not to eat or drink for 10 min after	Sedation, orthostasis, hypotension, dizziness, weight gain, EPS Advise patients about antipsychotic-	Decreased asenapine clearance with fluvoxamine.	

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	placing tablet under tongue	associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.		
quetiapine Seroquel, Seroquel XR, generics < \$30	Usual: 300 mg/day PO Maximum: 600 mg/day	Weight gain, sedation, orthostasis, metabolic disturbances. Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Ketoconazole dramatically increases quetiapine levels. Monitor for hypotension with antihypertensives.	First-line as monotherapy or in combination with lithium or divalproex.
Drug Class: Mood Stabilizers				
divalproex sodium Epival , Apo- Divalproex, other generics < \$30	750–2000 mg/day PO, adjusted according to serum levels (target range 350– 690 mcml/L)	Weight gain, thrombocytopenia, polycystic ovary syndrome, teratogenic, hyperammonemia (rare), hepatotoxicity (rare).	Avoid combining with carbamazepine. Use caution if combining with lamotrigine.	First-line as monotherapy or in combination with quetiapine or oral aripiprazole. On the NIOSH list of non- antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
				disposing of this drug. ^[b]
lamotrigine Lamictal , generics < \$30	100–300 mg/day PO	Rash common; severe rash rare. Slow titration reduces risk of severe dermatological reactions.	Divalproex doubles blood level of lamotrigine; lower initial and target doses are recommended.	
lithium  carbonate immediate-release Carbolith, Lithane, Apo-Lithium Carbonate, other generics < \$30	600–1800 mg/day PO, adjusted according to serum levels (target range 0.5–0.8 mmol/L; 0.5–0.8 mmol/L in elderly patients) Measure serum concentration 9–13 h post-dose	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment. Monitor thyroid and renal function at least every 6 months. Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term treatment with these agents is required. Avoid large changes in salt or coffee intake.	First-line as monotherapy or in combination with quetiapine or oral aripiprazole. Stop lithium temporarily during acute intermittent illnesses causing fluid and electrolyte losses.
lithium  carbonate controlled-release Lithmax \$30–60	600–1800 mg/day PO, adjusted according to serum levels (target range 0.5–0.8 mmol/L; 0.5–0.8 mmol/L in elderly patients)	Highly toxic in overdose. Measure electrolytes and do 24-h urine for ClCr when starting long-term treatment.	Toxic levels may result when adding NSAIDs , ACEIs , ARBs and especially thiazide diuretics. Reduce lithium dose and check serum level if long-term	First-line as monotherapy or in combination with quetiapine or oral aripiprazole. Stop lithium temporarily during acute

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
	Measure serum concentration 9–13 h post-dose	Monitor thyroid and renal function at least every 6 months.	treatment with these agents is required. Avoid large changes in salt or coffee intake.	intermittent illnesses causing fluid and electrolyte losses.
		Inform patients of signs and symptoms of toxicity such as ataxia, tremor, sedation or agitation, diarrhea, vomiting.		
valproic acid Depakene , Apo-Valproic, other generics < \$30	750–2000 mg/day PO, adjusted according to serum levels (target range 350–690 mcml/L)	Weight gain, thrombocytopenia, polycystic ovary syndrome, teratogenic, hyperammonemia (rare), hepatotoxicity (rare).	Avoid combining with carbamazepine. Use caution if combining with lamotrigine.	First-line as monotherapy or in combination with quetiapine or oral aripiprazole. On the NIOSH list of non-antineoplastic hazardous drugs; precautions must be used by health-care workers when handling and disposing of this drug. ^[b]

^[a] Cost of 30-day supply for mean dose unless otherwise specified; includes drug cost only.

^[b] For more information, see www.cdc.gov/niosh/docs/2016-161/.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; EPS = extrapyramidal symptoms; NSAID = nonsteroidal anti-inflammatory drug

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